

PROFESSIONAL EXPERIENCE

Head of Research

June 2026 — Present

Tutor Intelligence

Watertown, MA

- Lead robot learning research connecting policy training, data quality, and fleet deployment on Data Factory 1, a 100-robot bimanual manipulation fleet
- Led team improvements to full-rate MCAP logging, camera capture timestamps, and headless VR teleoperation; co-authored the RSS 2026 workshop paper on demonstration quality
- Moved research training from AWS to a 32-H100 allocation and experiment tracking to self-hosted MLflow
- Designed Rust middleware for synchronized sensing, control, logging, and policy execution; brought an initial version onto robot hardware with motion inhibited
- Set the research roadmap for policy evaluation and human-intervention learning, and defined hiring needs across robotics, infrastructure, and hardware

Advisor

May 2026 — Present

Tola Capital

Seattle, WA

- Technical advisor on robotics and AI investments

Research Lead

September 2023 — June 2026

Research Scientist

Dec 2022 — August 2023

RAI Institute (formerly Boston Dynamics AI Institute)

Cambridge, MA

- Led development of handheld force-based data collection system for robot training, capturing force, vision, and proprioception for in-the-wild demonstrations
- Built video prediction system for robot policies using internet-scale video as prior (early 2023); requires only limited real-world demonstrations
- Designed multimodal architectures combining diverse sensors with internet-scale priors for learned manipulation
- Developed gradient-free RL enabling online learning with non-differentiable semantic functions (U.S. Patent pending, May 2025)
- Created task definition framework that makes human demonstrations more learnable for robots
- Built ROS 2 interfaces and controllers for custom grippers using force/torque sensors
- Led co-design of the Koala handheld data-capture device and robotic gripper; co-authored Koala and BRIDGE (under review)
- Founded two research teams to scale innovations: Foundation Models (10+ researchers) and Capture (30 people)
- Built 320 H100 GPU infrastructure for large-scale training; created institute-wide policy execution framework
- Published at CoRL 2024 and IJCAI 2024; co-authored “Elephants Don’t Write Sonnets” (forthcoming MIT Press chapter)
- Established research partnerships with Google, Columbia, ETH Zurich, Agile Robots

Staff Software Engineer

August 2022 — October 2022

Righthand Robotics

Somerville, MA

- Worked on vision and planning team to improve pick and place performance for Rightpick systems
- Position ended due to company layoffs

CEO / Co-Founder, Odefi Inc.

Mar 2019 — Jan 2022

Columbia IBM Blockchain Accelerator

New York, NY

- Built liquidity automation system for MakerDAO network; architected smart contract infrastructure
- Led technical architecture, product development, and investor pitches as part of Columbia IBM Blockchain Accelerator

Research Fellow

Sep 2018 — July 2022

Army Research Lab

Aberdeen, MD

- Won 1st place MineRL BASALT Competition (NeurIPS 2021): developed a hierarchical agent combining learned and engineered components
- Developed mobile manipulation approaches without runtime localization (IROS 2022 Best Paper Finalist)
- Developed multimodal geometric learning fusing vision and tactile sensing for robotic grasping (ICRA 2019)
- Built simulation-to-real transfer pipelines for robotic grasping research

Engineering Intern

Jun 2016 — Aug 2016

Goldman Sachs — Margin Technology

New York, NY

- Built graph-based calculation prioritization system (Neo4j) optimizing ordering across 10K+ daily margin computations
- Developed Java/Angular.js system for dynamic prioritization weight adjustment

Engineering Intern

May 2015 — Aug 2015

Goldman Sachs — Valuations Technology

New York, NY

- Rebuilt FVA Gating Tool for derivatives valuations; implemented high-performance grid (100K+ rows, sub-second rendering)

Engineering Intern

Streakfire LLC

Jun 2014 — Sep 2014
Wayland, MA and Dorado, PR

- Produced ad campaign in Puerto Rico to promote technical accelerator for job creation
- Coordinated with local government to leverage publicity expertise and infrastructure

TECHNICAL EXPERIENCE

Koala and BRIDGE: Contact-Rich Manipulation

RAI Institute

2024 — 2026

Cambridge, MA

- Led work connecting handheld data capture, robotic gripper co-design, force sensing, and learned policy execution
- Co-authored BRIDGE, which combines handheld demonstrations with targeted teleoperation through state-gated policy experts
- Co-authored Koala, a jointly designed handheld and actuated gripper platform for dexterous manipulation learning

Theia: Vision Foundation Models for Robotics

RAI Institute, CoRL 2024

2024

Cambridge, MA

- Developed a spatial-token distillation framework combining diverse pretrained vision models into a compact representation for robot learning
- Evaluated representations on simulated and real-world manipulation tasks; released models and code

Multiple View Performers for Shape Completion

Robotics at Google, Army Research Lab, Columbia University

2021 — 2022

New York, NY

- Researched novel deep learning approach for multiple view completion without registering views
- Developed a process to leverage Performer attention layers developed by Google to encode multiple reconstruction images
- Released preprint in September 2022 (arXiv:2209.06291)

Data Driven Strand Simulation

Columbia University

Began Dec 2021

New York, NY

- Researched hair simulation algorithms to run in graph neural networks offering a 400% speedup
- Developed methods to compare python vs. C++ pytorch models
- Research collaboration on learned strand simulation

MineRL Basalt Competition

Neurips 2021

Jul 2021 — Dec 2021

New York, NY

- Researched the intersection of engineered and learned knowledge to develop an autonomous Minecraft agent using human demonstration data and won first place at MineRL Basalt at Neurips 2021 in collaboration with ARL and UMBC
- Researched learned visual navigation methods and developed a CNN state classifier using human-labeled data
- Work published at AAAI-Make 2022 and presented at Neurips 2021

Mobile Manipulation Leveraging Multiple Views

Columbia Robotics Lab

Jan 2020 — Oct 2022

New York, NY

- Researched deep learning approaches to mobile manipulation without localizing the robot at runtime
- Explored novel simulation based techniques for generating data with real-world scanned environments
- Improved previous navigation work via predicted panoramic target goals from nearby environment reconstruction
- Published to IROS 2022 and nominated for best paper in mobile manipulation

Learning from Electromyography Synergies to Grasp Novel Objects by Superquadric Representation

Columbia Robotics Lab

Jun 2018 — May 2020

New York, NY

- Collaborated with students to create a system for learning grasp synergies using a CTRL-Labs arm band and mapping the EMG signals to an anthropomorphic Seed robotic hand
- Helped develop an algorithm for planning and executing grasps using superquadric representations of novel objects with successful real-world demonstrations
- Work presented at Columbia Data Science Day 2019

Learning Your Way Without Map or Compass: Panoramic Target Driven Visual Navigation

Columbia Robotics Lab

Jan 2018 — Sep 2019

New York, NY

- Researched novel visual navigation methodology using RGBD panoramic target goals and behavioral cloning
- Developed a system architecture to embed images using an autoencoder and a policy model to control the robot
- Explored optimization strategies to develop training data from real-world environments without human intervention
- Work published to IROS 2020 and presented at NERC 2019

Multi-Modal Geometric Learning for Grasping and Manipulation

Columbia Robotics Lab

Sep 2017 — Sep 2019

New York, NY

- Incorporated tactile information to estimate shape geometry using vision and touch via deep learning
- Created a novel machine learned model for estimating shape via multi-modal information
- Work published to ICRA 2019 and to a RSS 2017 workshop

Human Robot Interface for Assistive Grasping

Columbia Robotics Lab and CUMC

Jan 2017 — Dec 2018

New York, NY

- Created a novel interface for enabling robot control for spinal cord injury patients using an sEMG device
- Benchmarked the sEMG interface against other modalities including an Amazon Echo and a toggle switch
- Work presented at Columbia Data Science Day 2018

Research in Bit Width Resolution

Columbia University

Jan 2016 — Dec 2016

New York, NY

- Worked with Professor Stephen Edwards at Columbia University to add Z3's SMT framework to an existing compiler project in order to resolve variable bit widths at compile time

Research in Shape Completion

Columbia Robotics Lab

Sep 2016 — May 2017

New York, NY

- Worked with Professor Peter Allen in the Columbia Robotics Lab at Columbia University to optimize an existing platform utilizing CUDA
- Evaluated the ability to utilize a semantic pre-processor to identify objects in a scene to be completed using the existing tool

Research in Data Visualization

Columbia University

Sep 2015 — Dec 2015

New York, NY

- Worked with Professor John Kender at Columbia University to provide visualization of the correlation between visual and textual memes in online video data and provided an analysis on the most effective ways of visualizing co-clustered data

Research in the Production of Litecoin ASICs

Columbia University

Jan 2014 — May 2014

New York, NY

- Cooperated with Professor Simha Sethumadhavan on the feasibility of producing Litecoin Mining ASICs
- Independently designed all of the ASIC schematics, performed cost-benefit analysis of the ASIC and maintained knowledge on which crypto-currencies were most profitable at any time

Written Values Affirmation Intervention to Identify the Unique Linguistic Features of Stigmatized Groups

LIRSM

Sep 2014 — Jan 2015

New York, NY

- Responsible for developing web application in Node.js and Mongo to easily add and retrieve information from csv files
- Participated in IT work and assisted with sessions in informing individuals on using UNIX
- Developed strategies to acquire study data more efficiently and help audit costs on services

Walking in Their Shoes: Poverty in America

Yale

Sep 2014 — Jan 2015

New Haven, CT

- Developed interface and game for flexible assignment of agency to test subjects to explore the impact that games have on depictions of poverty
- Worked with MongoDB and Angular.js to create flexible tiled gameplay

PUBLICATIONS

1. Hajj-Ahmad, Amar, Zubin Kremer Guha, Tim Fofonoff, Zhi Ern Teoh, Ciarán T. O'Neill, Ben Thacher, Igor Fala, Vidullan Surendran, Murphy Wonsick, Peter Whitney, and **David Watkins**. "Koala Gripper: Co-designing Robotic Grippers and Data-Capture Devices for Scaling Dexterous Manipulation Learning." arXiv:2608.20546, 2026. Under review at IEEE Transactions on Robotics (T-RO). <https://arxiv.org/abs/2608.20546>
2. Surendran, Vidullan, Neehar Peri, and **David Watkins**. "Bridging Handheld and Teleoperated Supervision for Contact-Rich Manipulation via State-Gated Experts." arXiv:2606.26603, 2026. Under review at ICRA 2027. <https://arxiv.org/abs/2606.26603>
3. Kapoor, Kunal, Shiraz Khan, Joseph McCalmon, Jesse Michel, Arif Mohammed, Katerina Nikiforova, Adam Oppenheimer, Victor Szabo, and **David Watkins**. "Maintaining Demonstration Quality in a 100-Robot Teleoperation Pipeline." Workshop on "It's the Demos: A Deep Look at the Role of Demonstration Quality in Imitation-Based Robot Manipulation," Robotics: Science and Systems (RSS), Sydney, Australia, 2026.
4. Tellex, Stefanie, and **David Watkins**. "Elephants Don't Write Sonnets: The Grounded Turing Test for Embodied AI." In *Designing an Intelligence*, edited by George Konidaris. MIT Press, forthcoming (2026 planned). <https://h2r.cs.brown.edu/wp-content/uploads/tellexwatkins2026.pdf>
5. Shang, Jinghuan, Karl Schmeckpeper, Brandon B. May, Maria Vittoria Minniti, Tarik Kelestemur, **David Watkins**, and Laura Herlant. 'Theia: Distilling Diverse Vision Foundation Models for Robot Learning'. In 8th Annual Conference on Robot Learning, 2024. <https://openreview.net/forum?id=yLZHv1wUcI>.
6. Cohen, Vanya, Jason Xinyu Liu, Raymond Mooney, Stefanie Tellex, and **David Watkins**. "A Survey of Robotic Language Grounding: Tradeoffs Between Symbols and Embeddings." Proceedings of IJCAI 2024, Survey Track, pp. 7999–8009.

7. Novoseller, Ellen, Vinicius G. Goecks, **David Watkins**, Josh Miller, and Nicholas Waytowich. "DIP-RL: Demonstration-Inferred Preference Learning in Minecraft." ICML 2023 Workshop on The Many Facets of Preference Learning.
<https://arxiv.org/abs/2307.12158>
8. Milani, S., Kanervisto, A., Ramanauskas, K., Schulhoff, S., Houghton, B., Mohanty, S., ... & Shah, R. (2023). Towards Solving Fuzzy Tasks with Human Feedback: A Retrospective of the MineRL BASALT 2022 Competition. arXiv preprint arXiv:2303.13512.
9. Choromanski, K.M., Sehanobish, A., Lin, H., Zhao, Y., Berger, E., Parshakova, T., Pan, A., **Watkins, D.**, Zhang, T., Likhoshesterov, V. and Chowdhury, S.B.R., 2023, July. Efficient graph field integrators meet point clouds. In International Conference on Machine Learning (pp. 5978-6004). PMLR.
10. Hu, Jiaheng, **David Watkins**, and Peter Allen. "Teleoperated Robot Grasping in Virtual Reality Spaces." arXiv preprint arXiv:2301.13064 (2023).
11. **Watkins-Valls, D.**, Allen, P., Choromanski, K., Varley, J., & Waytowich, N. (2022). Multiple View Performers for Shape Completion. arXiv preprint arXiv:2209.06291.
12. **Watkins, David Joseph**. (2022). Learning Mobile Manipulation. Columbia University.
<https://doi.org/10.7916/V9YM-TQ84>
13. **Watkins-Valls, D.**, Maia H., Varley J., Seshadri M., Sanabria J., Waytowich, N., & Allen, P. (2022). Mobile Manipulation Leveraging Multiple Views. 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS 2022
14. Shah, Rohin, Steven H. Wang, Cody Wild, Stephanie Milani, Anssi Kanervisto, Vinicius G. Goecks, Nicholas Waytowich, **David Watkins-Valls**, Bharat Prakash, Edmund Mills, Divyansh Garg, Alexander Fries, Alexandra Souly, Jun Shern Chan, Daniel del Castillo, and Tom Lieberum. "Retrospective on the 2021 MineRL BASALT Competition on Learning from Human Feedback." NeurIPS 2021 Competitions and Demonstrations Track, PMLR 176:259–272, 2022.
<https://proceedings.mlr.press/v176/shah22a.html>
15. Goecks, Vinicius G., et al. "Combining Learning from Human Feedback and Knowledge Engineering to Solve Hierarchical Tasks in Minecraft." ArXiv:2112.03482 [Cs], Dec. 2021. arXiv.org, <http://arxiv.org/abs/2112.03482>. Accepted to AAAI-Make 2022.
16. **Watkins-Valls, D.**, Xu, J., Waytowich, N., & Allen, P. (2020). Learning your way without map or compass: Panoramic target driven visual navigation. 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS 2020
17. Akinola, Ireteayo, Zizhao Wang, Junyao Shi, Xiaomin He, Pawan Lapborisuth, Jingxi Xu, **David Watkins-Valls**, Paul Sajda, and Peter Allen. "Accelerated Robot Learning via Human Brain Signals." In 2020 IEEE International Conference on Robotics and Automation (ICRA), pp. 3799-3805. IEEE, 2020.
18. Wu, B., Akinola, I., Gupta, A., Xu, F., Varley, J., **Watkins-Valls, D.**, & Allen, P. K. (2020). Generative Attention Learning: a "GenerAL" framework for high-performance multi-fingered grasping in clutter. Autonomous Robots, 1-20.
19. **Watkins, David**, Chaiwen Chou, Caroline Weinberg, Jacob Varley, Kenneth Lyons, Sanjay Joshi, Lynne Weber, Joel Stein, and Peter Allen. "Human Robot Interface for Assistive Grasping." arXiv:1804.02462, 2018. <https://arxiv.org/abs/1804.02462>
20. **Watkins-Valls, D.**, Varley, J. & Allen, P. Multi-Modal Geometric Learning for Grasping and Manipulation. 2019 IEEE International Conference on Robotics and Automation (ICRA). IEEE, 2019.
21. Abhi Gupta, Jingya Bi, Ashwin Jayaraman, Max Xu, **David Watkins** and Professor Peter Allen. "Learning from Electromyography Synergies to Grasp Novel Objects by Super Quadric Representation (Poster)" In: Columbia Data Science Day (2019).
22. Jacob Varley, **David Watkins-Valls**, and Peter Allen. "Multi-Modal Geometric Learning for Grasping and Manipulation (Poster)". In: Columbia Data Science Day (2018).
23. Jacob Varley, **David Watkins**, and Peter Allen. "Visual-Tactile Geometric Reasoning (Abstract and Poster)". In: Data-Driven Manipulation workshop, Robotics: Science and Systems (2017).
24. **David Watkins-Valls**, Chaiwen Chou, Caroline Weinberg, Jacob Varley, Lynne Weber, Adam Blanchard, Peter Allen, Joel Stein "Human Robot Interface for Assistive Grasping (Poster)". In: New England Manipulation Symposium (2017).
25. **Watkins-Valls, David**, and Peter Allen. "Optimizing the Shape Completion Pipeline." Columbia University, 2017. Technical report.
26. **David Watkins-Valls** "Script Mining With ASICs" (2014).

1. **Watkins, David**, and Stefanie Tellex. "A Theory of Mind for Robotics." *What to Tell the Robot*, October 2, 2026. <https://whattotelltherobot.com/p/a-theory-of-mind-for-robotics>
2. Tellex, Stefanie, and **David Watkins**. "The Law of Leaky Abstractions and Robotics." *What to Tell the Robot*, September 25, 2026. <https://whattotelltherobot.com/p/the-law-of-leaky-abstractions-and>
3. Tellex, Stefanie, and **David Watkins**. "Robotics Really Is Harder." *What to Tell the Robot*, September 18, 2026. <https://whattotelltherobot.com/p/robotics-really-is-harder>
4. **Watkins, David**, and Stefanie Tellex. "The Bottleneck in Robotics Is the Reward, Not the Policy." *What to Tell the Robot*, September 15, 2026. <https://whattotelltherobot.com/p/a-million-rewards-a-second>
5. **Watkins, David**, and Stefanie Tellex. "The Reward Is In There." *What to Tell the Robot*, September 1, 2026. <https://whattotelltherobot.com/p/the-reward-is-in-there>
6. **Watkins, David**, and Stefanie Tellex. "Diffusion Policies are Fancy Lookup Tables." *What to Tell the Robot*, July 21, 2026. <https://whattotelltherobot.com/p/diffusion-policies-are-fancy-lookup>
7. Tellex, Stefanie, and **David Watkins**. "We Got a New Nine." *What to Tell the Robot*, July 11, 2026. <https://whattotelltherobot.com/p/we-got-a-new-nine>
8. Tellex, Stefanie, and **David Watkins**. "I Need Space." *What to Tell the Robot*, July 6, 2026. <https://whattotelltherobot.com/p/i-need-space>
9. Tellex, Stefanie, and **David Watkins**. "Why Robotics is a Pre-Paradigm Field." *What to Tell the Robot*, June 6, 2026. <https://whattotelltherobot.com/p/why-robotics-is-a-pre-paradigm-field>
10. Tellex, Stefanie, and **David Watkins**. "I Got Sycophanted." *What to Tell the Robot*, May 20, 2026. <https://whattotelltherobot.com/p/i-got-sycophanted>
11. Tellex, Stefanie, and **David Watkins**. "How to Build Safe AI (Without Making the AI Safe)." *What to Tell the Robot*, May 13, 2026. <https://whattotelltherobot.com/p/how-to-build-safe-ai-without-making>
12. **Watkins, David**. "Getting a Grip on Robotic Data Collection." *RAI Institute Blog*, May 12, 2026. <https://rai-inst.com/resources/blog/handheld-robotic-data-collection/>
13. Tellex, Stefanie, and **David Watkins**. "The First Hit is Free." *What to Tell the Robot*, May 7, 2026. <https://whattotelltherobot.com/p/the-first-hit-is-free>
14. **Watkins, David**, and Stefanie Tellex. "What Babies Know That Robots Don't." *What to Tell the Robot*, April 15, 2026. <https://whattotelltherobot.com/p/what-babies-know-that-robots-dont>
15. **Watkins, David**, and Stefanie Tellex. "Consciousoids." *What to Tell the Robot*, March 30, 2026. <https://whattotelltherobot.com/p/consciousoids>
16. Tellex, Stefanie, and **David Watkins**. "Elephants Still Don't Play Chess." *What to Tell the Robot*, March 2, 2026. <https://whattotelltherobot.com/p/elephants-still-dont-play-chess>
17. Tellex, Stefanie, and **David Watkins**. "Radiate Love: a 2025 Retrospective." *What to Tell the Robot*, February 3, 2026. <https://whattotelltherobot.com/p/radiate-love-a-2025-retrospective>
18. **Watkins, David**, and Stefanie Tellex. "Satisfying a 31-Year-Old Karaoke Dream." *What to Tell the Robot*, January 16, 2026. <https://whattotelltherobot.com/p/satisfying-a-31-year-old-karaoke>
19. Tellex, Stefanie, and **David Watkins**. "Where the Curves Cross." *What to Tell the Robot*, January 9, 2026. <https://whattotelltherobot.com/p/where-the-curves-cross>
20. **Watkins, David**. "2025 in Review: The Year of Showing Up." *What to Tell the Robot*, December 31, 2025. <https://whattotelltherobot.com/p/2025-in-review-the-year-of-showing>
21. **Watkins, David**, and Stefanie Tellex. "The Deer Island Marvel." *What to Tell the Robot*, December 5, 2025. <https://whattotelltherobot.com/p/the-deer-island-marvel>
22. Tellex, Stefanie, and **David Watkins**. "Qualitative Simulation of Swine Production." *What to Tell the Robot*, November 19, 2025. <https://whattotelltherobot.com/p/qualitative-simulation-of-swine-production>

23. Tellex, Stefanie, and **David Watkins**. "The Grounded Turing Test." *What to Tell the Robot*, September 2, 2025.
<https://whattotelltherobot.com/p/the-grounded-turing-test>
24. **Watkins, David**, and Stefanie Tellex. "From Physics to Embodied Intelligence." *What to Tell the Robot*, August 27, 2025.
<https://whattotelltherobot.com/p/from-physics-to-embodied-intelligence>
25. Tellex, Stefanie, and **David Watkins**. "Taking a Quadruped Robot to School." *What to Tell the Robot*, August 20, 2025.
<https://whattotelltherobot.com/p/taking-a-quadruped-robot-to-school>
26. Tellex, Stefanie, and **David Watkins**. "What to Tell the Robot." *What to Tell the Robot*, July 18, 2025.
<https://whattotelltherobot.com/p/what-to-tell-the-robot>

EDUCATION

PhD in Computer Science , <i>Columbia University</i>	Sep 2017 — May 2022
Advisor: <i>Prof. Peter Allen</i> , Thesis: <i>Learning Mobile Manipulation</i>	
<i>Army Research Lab Research Fellow</i>	Sep 2018 — July 2022
MPhil in Computer Science , <i>Columbia University</i>	Sep 2017 — May 2019
MS in Computer Science , <i>Columbia University</i> , 4.0 GPA	Sep 2016 — May 2017
<i>CA Fellowship</i>	Sep 2016 — Jan 2017
BS in Computer Science , <i>Columbia University</i> , 3.7 GPA	Sep 2012 — May 2016
Marian High School	Oct 2010 — May 2012
<i>Class President</i>	
<i>Salutatorian</i>	
<i>National Honors Society</i>	

AWARDS

MineRL BASALT Competition: 1st Place and Most Human-Like Agent , <i>NeurIPS 2021</i>	Dec 2021
<i>Team result for combining learning from human feedback with knowledge engineering in Minecraft</i>	
Best Paper in Mobile Manipulation Finalist , <i>Kyoto, Japan</i>	Oct 2022
<i>Selected for my work titled Mobile Manipulation Leveraging Multiple Views submitted to IROS 2022</i>	
International House Member , <i>New York, NY</i>	Sep 2016 — May 2018
<i>Competitively selected scholars and young professionals from around the world who are challenged to become globally-minded leaders</i>	
CA Fellowship , <i>Columbia University</i>	Sep 2016 — Dec 2016
<i>MS students who have proven themselves to be exceptional will receive paid tuition and stipend</i>	
Goldman Sachs Code Golf Champion , <i>New York, NY</i>	Aug 2016
<i>Awarded for shortest possible source code that implements a certain algorithm</i>	
Dean's List , <i>Columbia University</i>	Spring 2013, Spring 2015, Spring 2016
<i>A list of students recognized for academic achievement during a semester by the dean of the college they attend</i>	
Residential Incubator Fellow , <i>Columbia University</i>	Sep 2012 — May 2014
<i>Students interested in entrepreneurship participating in a student incubator</i>	

ACTIVITIES

Tactile Sensing for Robotic Foundation Models Workshop Co-Organizer, RSS	2026
New England Manipulation Symposium (NEMS) Co-Organizer	2025
IJRR Reviewer	2023
NSF Reviewer	2023
ICRA Paper Reviewer	2022, 2025
Computer Science Student Faculty Representative	2018 — 2020
IEEE RA-L Paper Reviewer	2021, 2024, 2025
IROS Paper Reviewer	2018, 2021, 2023, 2024
CoRL Paper Reviewer	2020, 2024

SKILLS

Languages	Python, C++, Rust, CUDA, Bash, $\text{\LaTeX}$, SQL
ML / AI	Foundation Models, Transformers, Diffusion Policies, Imitation Learning, RLHF, Distributed Training
Infrastructure	GPU Clusters, Docker, Kubernetes, MLflow, Git, CI/CD
Robotics	Manipulation, Navigation, SLAM, ROS 2, PyBullet, MuJoCo, Isaac
Leadership	Research Strategy, Team Building, Hiring, Technical Partnerships
Communication	English, Spanish

VOLUNTEER WORK

Multisensory Reading Centers of Puerto Rico , San Juan, PR	Sep 2017 — Present
<i>Over 150 hours of volunteer service by performing IT help to provide access to effective literacy instruction for struggling readers</i>	
Watkins-Valls Family Foundation , Boston, MA	Sep 2013 — Present
<i>Providing scholarships and academic support to underprivileged students in Massachusetts, New York, and Puerto Rico</i>	
Pine St. Inn , Boston, MA	Dec 2010 — May 2012
<i>Over 25 hours of volunteer service at a soup kitchen for the homeless</i>	
Sisters of St. Joseph , Cambridge, MA	Dec 2010 — May 2012
<i>Over 25 hours of volunteer service through entertaining and assisting retired nuns</i>	

PRESENTATIONS

What Data Robots Need: From Co-Designed Handheld Capture to Fleet-Scale Flywheels , Extropic, Waltham, MA	Jul 2026
<i>Invited talk on robot data collection, from co-designed handheld force-based capture to fleet-scale data flywheels, presented to the Extropic team</i>	
Tactile Sensing for Robotic Foundation Models Workshop Co-Organizer , RSS 2026, Sydney	Jul 2026
<i>Co-organized workshop on the role of tactile sensing in robotic foundation models and large-scale data collection</i>	
What Data Robots Need: From Co-Designed Handheld Capture to Fleet-Scale Flywheels , RSS 2026 Workshop (It's the Demos), Sydney	Jul 2026
<i>Invited workshop talk tracing robot data collection from co-designed handheld force-based capture to fleet-scale data flywheels, drawing on work at the RAI Institute and Data Factory 1 (DF1)</i>	
Why Robots? Data-Driven Manipulation for the World's Hardest Sorting Problem , Brown University	Feb 2026
<i>Guest lecture in Stefanie Tellex's class on robotic manipulation for landfill restoration, covering multimodal sensing, force-based data collection, and the capability gap between warehouse and unstructured-environment robots</i>	
New England Manipulation Symposium (NEMS) 2025 Co-Organizer , MIT	Apr 2025
<i>Co-organized conference with Lael Odhner and Kaitlyn Becker; coordinated paper acceptances, speaker scheduling, and venue logistics</i>	
The Future of Intelligent Robotics , Dr. Waku YouTube Interview	2025
<i>Featured interview discussing robotics' "ChatGPT moment" and the future of embodied AI</i>	
Lions in AI Panel , Columbia Alumni Association of Boston	2025
<i>Panel discussion on AI and robotics for Columbia alumni community</i>	
GTC 2025 Summary , RAI Institute	Mar 2025
<i>Presented key takeaways from NVIDIA GPU Technology Conference to institute leadership</i>	
Theia: Distilling Diverse Vision Foundation Models , CoRL 2024, Munich	Nov 2024
<i>Published work on foundation model distillation for robot learning at Conference on Robot Learning</i>	
Foundation Models for Robotics , Duke University	Feb 2024
<i>Presentation on state of the art foundation models for robotic learning</i>	
Helios: High Dimensional Predictive Foundation Models as Knowledge Transfer Systems , Brown University	Jul 2023
<i>Building large scale video predictive foundation models for robotics at the AI Institute</i>	
From One to Many: How to leverage multiple views in shape completion , Robotics at Google	Dec 2022
<i>A discussion on how to leverage multiple views for shape completion presented to researchers at Robotics at Google</i>	
Learning Mobile Manipulation , Harvard - Harvard Biorobotics Laboratory	Nov 2022
<i>Presented an abbreviated version of my thesis enabling mobile robots to manipulate objects to professor Robert Howe's lab</i>	
Learning Mobile Manipulation , Tufts - Human Robotics Interaction Laboratory	Nov 2022
<i>Presented an abbreviated version of my thesis enabling mobile robots to manipulate objects to professor Matthias Scheutz's lab</i>	
Learning Mobile Manipulation , MIT - Improbable AI Lab	Nov 2022
<i>Presented an abbreviated version of my thesis enabling mobile robots to manipulate objects to professor Pulkit Agrawal's lab</i>	
Learning Mobile Manipulation , Boston Dynamics	Nov 2022
<i>Presented an abbreviated version of my thesis enabling mobile robots to manipulate objects</i>	
Learning Mobile Manipulation , Boston Dynamics AI Institute	Nov 2022
<i>Presented a collection of works done during my Ph.D.</i>	
Learning Mobile Manipulation , MITRE	Nov 2022
<i>Presented an abbreviated version of my thesis enabling mobile robots to manipulate objects</i>	
Learning Mobile Manipulation , Rotor.ai	Oct 2022
<i>Presented an abbreviated version of my thesis enabling mobile robots to manipulate objects</i>	
Learning Mobile Manipulation , STR	Oct 2022
<i>Presented an abbreviated version of my thesis enabling mobile robots to manipulate objects</i>	

Mobile Manipulation Leveraging Multiple Views, IROS 2022 <i>Presented work on localization free mobile manipulation that was nominated for best paper in mobile manipulation</i>	Oct 2022
P.h.D. Defense: Learning Mobile Manipulation, Columbia University <i>Presented my dissertation defense enabling mobile robots to manipulate objects</i>	May 2022
Minecraft Basalt Interview, Yannic Kilcher YT <i>Interviewed on recent first place win in Minecraft Basalt RL competition.</i>	Jan 2022
Learning Mobile Manipulation, ROAM Lab <i>Presented work done in preparation of defense of my thesis to the ROAM lab at Columbia University.</i>	Dec 2021
Learning Mobile Manipulation, CAIR Lab <i>Presented my thesis proposal to the CAIR lab at Columbia University.</i>	Feb 2021
IROS 2020: Learning Your Way Without Map or Compass, IROS 2020 <i>Presented my Learning Your Way work at IROS 2020 via the online conference.</i>	Oct 2020
Demystifying the Dissertation, Columbia University <i>Presented my thesis proposal and help current graduate students understand the process of completing a PhD.</i>	Jun 2020
Thesis Proposal: Learning Mobile Manipulation, New York, NY <i>Presented and defended my thesis proposal in learning mobile manipulation in January 2020.</i>	Jan 2020
Research Talk, Harlem Children's Zone STEM Exposure <i>Shared my research as part of an initiative to help teach children in Harlem methods in STEM research.</i>	Nov 2019
Learning Your Way Talk, NYU Reading Group <i>Presented Learning Your Way work to the NYU robotics reading group.</i>	Oct 2019
Learning Your Way Talk, NERC 2019 <i>Presented Learning Your Way work in front of audience of robotics researchers at NERC 2019 conference.</i>	Oct 2019
Visual Tactile Completion, Emptor Lightning Talks <i>Presented work on visual tactile grasping as well as next steps as part of Emptor's lightning talks.</i>	Sep 2019
Odefi Pitch, Columbia IBM Blockchain Accelerator Demo Day 2019 <i>Enabling credit default swaps on the ethereum network pitched to investors at the capstone event for the Columbia IBM Blockchain Accelerator.</i>	May 2019
Visual Tactile Grasping, Samsung Research NYC <i>Presented work on visual tactile grasping as well as next steps.</i>	July 2019
Candidacy Exam: Simulation for Real World Robotics, New York, NY <i>A high-level overview of how real-world robotics can be enabled through simulation.</i>	May 2019
Using Simulation to Enable Generated Art and Robotics, Making Art in the Age of Algorithms <i>A high-level overview of how robotics can be enabled through simulation as part of a series of lightning talks about art and algorithms.</i>	Dec 2018
Visual Tactile Completion Poster, Data Science Day 2018 <i>This work provides an architecture that incorporates depth and tactile information to create rich and accurate 3D models useful for robotic manipulation tasks presented at Data Science Day 2018 at Columbia University.</i>	Mar 2018
Lecture: ROS Tutorial, New York, NY <i>An introductory tutorial on ROS and use of robotics in the Columbia Robotics Lab to aspiring roboticists.</i>	Jan 2018
Providing Context to Startup Culture, New York, NY <i>An analysis on the effectiveness of a startup based on the type of culture it maintains as well as effects on profit/loss.</i>	May 2016

MEMBERSHIPS

ACM	2016 — Present
IEEE	2016 — Present
SHPE	2018 — Present
AAAI	2022 — Present

TEACHING EXPERIENCE

Humanoid Robotics (COMSW 6731), Teaching Assistant, Graduate Level	Spring 2018
Computational Aspects of Robotics (COMSW 4733), Teaching Assistant, Graduate Level	Fall 2017
Programming Languages and Translators (COMSW 4115), Teaching Assistant, Graduate Level	Fall 2016
Object Oriented Programming and Design in Java (COMS 1007), Teaching Assistant, Undergraduate Level	Fall 2016
Programming Languages and Translators (COMSW 4115), Teaching Assistant, Graduate Level	Spring 2016
Object Oriented Programming and Design in Java (COMS1007), Teaching Assistant, Undergraduate Level	Fall 2014
Fundamentals of Computer Systems (CSEE3827), Teaching Assistant, Undergraduate Level	Spring 2014
Object Oriented Programming and Design in Java (COMS1007), Teaching Assistant, Undergraduate Level	Fall 2013

REFERENCES

References available upon request.